Sec 12.4 Equilibrium Temperature Distribution(steady state)

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Of interest in many physical problems is what we call the equilibrium state or steady state. This is the state where a system stops undergoing change.

Let's apply this to the 1-d heat equation. $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} + Q$ with $Q = 0$

If we are in equilibrium this means that the temperature is no longer changing or that $\frac{\partial u}{\partial t} = 0$

Thus we have the now ODE

$$\frac{d^2u}{dx^2} = 0$$

We will still have an initial condition
$u(x, 0) = f(x)$

Whether or not we will need the initial condition will depend on out boundary conditions.

Let's first consider Dirichlet boundary conditions(prescribed temperature at the ends)

$$u(0, t) = T_1 \text{ and } u(L, t) = T_2$$

We then want to solve the ODE : $\frac{\partial^2 u}{\partial x^2} = 0$ with boundary conditions $u(0, t) = T_1$ and $u(L, t) = T_2$

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Insulated boundaries:

Let' solve the same problem but with insulated end boundary conditions. BCs: $\frac{\partial u}{\partial x}(0) = 0 \ \ and \frac{\partial u}{\partial x}(L) = 0$

With an initial condition of $u(x, 0) = f(x)$

First let's think about it…



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Homework example.

From sec 12.4 Problem 1 part (f)

Determine the equilibrium temperature distribution for a one dimensional rod with the following sources and boundary conditions.

$$\frac{Q}{K_0} = x^2 , \qquad u(0) = T_1 , \qquad \frac{\partial u}{\partial x}(L) = 0$$

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Home Work Example

From Sec 12.4 Problem 4

If both ends of a rod are insulated show, using the partial differential equation , that the total thermal energy in the rod is constant.